

ShiftGuard10: Distribution-Shift Robust Image Classification via WideResNet-40-10 with 3-Seed Ensemble

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I. ABSTRACT

This paper presents ShiftGuard10 v7, a distribution-shift robust image classifier built on a WideResNet-40-10 backbone ($\sim 54\text{M}$ parameters). The system employs a 3-seed ensemble trained over 300 epochs with Stochastic Weight Averaging (SWA), BalancedSoftmaxLoss, a rich augmentation pipeline (AugMix, GreyCutout, RandomErasing, GaussianNoise, CutMix, MixUp), and 30-round Test-Time Augmentation (TTA). The dataset contains 29,400 images across 10 classes with severe imbalance (5,000 to 100 samples per class). The best single-seed model achieves a macro F1-score of **0.9206** on the held-out validation set; the ensemble targets $F1 > 0.95$ on the test set.

II. INTRODUCTION

Image classification under real-world conditions frequently involves *distribution shift*—systematic differences between training and test images—and *class imbalance*, where some categories have orders of magnitude more labeled examples than others. ShiftGuard10 addresses both challenges on a 10-class CIFAR-style dataset containing 29,400 training images with class sizes ranging from 5,000 to 100 samples per class.

The system runs on a Tesla T4 GPU (15.6 GB VRAM) with Automatic Mixed Precision (AMP) enabled. Training uses three independent random seeds to produce an ensemble of three WideResNet-40-10 models, each trained for 300 epochs. At inference, 30-round TTA per model is applied, totalling 90 forward passes per test image. The objective is to maximise macro-averaged F1 score on the held-out test set, explicitly targeting $F1 > 0.95$ despite severe class imbalance and test-time distribution shift.

III. DATASET

The dataset contains 29,400 labeled RGB images (32×32 pixels) across 10 classes: *airplane*, *automobile*, *bird*, *cat*, *deer*, *dog*, *frog*, *horse*, *ship*, and *truck*. A stratified 95/5 split yields 27,930 training and 1,470 validation samples. Class distribution is severely imbalanced, with a 50:1 ratio between the most and least frequent class (Table 1).

A sqrt-inverse frequency weighted sampler addresses imbalance during training, with per-class weights $w_i = 1/\sqrt{N_i}$, providing balanced mini-batches without extreme oversampling of very rare classes.

TABLE 1
CLASS-WISE DATASET DISTRIBUTION

Class	Train	Val
airplane	4,750	250
automobile	4,750	250
bird	4,750	250
cat	4,750	250
deer	3,800	200
dog	3,800	200
frog	475	25
horse	475	25
ship	285	15
truck	95	5

IV. ARCHITECTURE

A. WideResNet-40-10

The backbone is a pre-activation Wide Residual Network (WRN-40-10) [1] with depth 40, width factor 10, and $\sim 54\text{M}$ parameters. The channel progression through three stages is $16 \rightarrow 160 \rightarrow 320 \rightarrow 640$. Each WideBlock applies $\text{BN} \rightarrow \text{ReLU} \rightarrow \text{Conv}_{3 \times 3} \rightarrow \text{Dropout}(p=0.3) \rightarrow \text{Conv}_{3 \times 3}$ with a 1×1 projection shortcut when spatial or channel dimensions change. Global average pooling feeds a single linear classifier head. Weights are initialised with Kaiming Normal (Conv), Xavier Normal (Linear), and ones/zeros (BN). The model is optionally compiled with `torch.compile()`.

B. BalancedSoftmaxLoss

To handle class imbalance at the loss level, BalancedSoftmaxLoss [2] adds $\log(\text{class_frequency})$ to logits before cross-entropy, shifting the decision boundary toward rare classes:

$$\tilde{\ell}_c = \ell_c + \log\left(\frac{N_c}{\sum_j N_j}\right).$$

Label smoothing ($\epsilon=0.1$) reduces overconfidence. The log-frequency biases for the 10 classes are: $[-1.77, -1.77, -1.77, -1.77, -2.00, -2.00, -4.07, -4.07, -4.59, -5.6]$.

V. TRAINING PIPELINE

A. Data Augmentation

A multi-layer augmentation pipeline addresses distribution shift and overfitting (Table 2). The key domain-specific

TABLE 2
AUGMENTATION PIPELINE

Augmentation	Configuration
AugMix	Stochastic chain mixing; CIFAR policy
AutoAugment	Optimised CIFAR-10 policy
ColorJitter	Brightness, contrast, saturation, hue
GreyCutout	5–12 px grey patch, $p=0.5$
RandomErasing	Random rectangle erasure
GaussianNoise	Additive noise injection
CutMix	$\alpha=1.0$, 25% per batch
MixUp	$\alpha=0.4$, 25% per batch

component is *GreyCutout*: it replicates grey square artifacts (5–12 px) observed in the dataset by filling a random patch with the correctly-normalised grey value $(0.5 - \mu_c) / \sigma_c \approx [0.035, 0.073, 0.205]$ per channel, directly teaching the model to handle these test-time corruptions. Size range 5–12 px slightly exceeds the observed 5–10 px range to add margin.

CutMix [4] pastes a rectangular crop from a shuffled image with labels mixed proportionally by area. MixUp [5] linearly interpolates two images and their label distributions. Both soft-label variants apply the same BalancedSoftmax log-frequency bias via a custom `soft_ce()` function. Each batch applies CutMix (25%), MixUp (25%), or standard BalancedSoftmaxLoss (50%).

B. Optimiser and LR Schedule

SGD with Nesterov momentum is used (LR = 0.1, momentum = 0.9, weight decay = 5×10^{-4}). A 5-epoch linear warmup is followed by cosine annealing to 0 over epochs 5–300. Batch size is 256 with 4 data-loading workers, pin memory, and persistent workers. Gradient norms are clipped at 1.0. Automatic Mixed Precision (AMP) via GradScaler is enabled throughout.

C. Stochastic Weight Averaging (SWA)

SWA [3] activates at epoch 225 (75% of training), averaging model weights over the final 75 epochs with a cyclic SWALR schedule (`swa_lr` = 0.01, `anneal_epochs` = 10). Batch-Norm statistics are recalculated on the full training set after weight averaging. A full checkpoint is saved after every epoch, enabling automatic resumption after interruption.

D. Test-Time Augmentation (TTA)

At inference each model performs 30 stochastic TTA forward passes using the full augmentation transform, plus 1 clean pass weighted $\times 3$ to anchor predictions:

$$P_{\text{model}} = \frac{\sum_{i=1}^{30} p_i^{\text{TTA}} + 3 p^{\text{clean}}}{33}.$$

The 3 model outputs are then averaged with equal weight to form the final ensemble prediction, combining 90 total forward passes per test image.

TABLE 3
TRAINING CONFIGURATION AND PERFORMANCE

Parameter / Metric	Value
Architecture	WideResNet-40-10
Parameters	~54M
Training epochs	300
Batch size	256
Optimiser	SGD (Nesterov)
Base LR	0.1 (cosine anneal $\rightarrow$ 0)
Weight decay	5×10^{-4}
SWA start epoch	225
Label smoothing	0.1
Dropout	0.3 (per block)
Seeds	{23, 71, 314}
TTA rounds	30 per model
Val F1 (seed 23)	0.9206
Ensemble target	F1 > 0.95

VI. RESULTS

The best single-seed model (seed 23) achieves a macro F1-score of **0.9206** on the held-out 1,470-sample validation set. The full 3-seed ensemble with 30-round TTA targets F1 > 0.95 on the test set. Table 3 summarises training configuration and performance metrics.

VII. DISCUSSION

ShiftGuard10 demonstrates that layering complementary robustness strategies yields strong performance under severe imbalance and distribution shift. BalancedSoftmaxLoss shifts prediction mass toward underrepresented classes (e.g., *truck* with only 95 training samples). GreyCutout directly targets a domain-specific artifact, improving test-time robustness. SWA finds a flatter loss landscape than the final SGD iterate; combined with cosine annealing, it produces better-calibrated outputs [3].

The 3-seed ensemble reduces prediction variance arising from stochastic weight initialisation and augmentation ordering, while 30-round TTA smooths image-level randomness at test time. Together these strategies are expected to close the gap between the single-seed F1 of 0.9206 and the 0.95 target.

VIII. CONCLUSION

This paper presented ShiftGuard10 v7, a WideResNet-40-10 ensemble system for robust image classification under distribution shift and class imbalance. The system achieves macro F1 = 0.9206 with a single model and targets F1 > 0.95 with a 3-seed ensemble and 30-round TTA. The training pipeline integrates BalancedSoftmaxLoss, SWA, AugMix, GreyCutout, CutMix, and MixUp, each addressing a specific failure mode. Future work will explore larger backbones, extended ensemble sizes, and adaptive TTA strategies.

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REFERENCES

- [1] S. Zagoruyko and N. Komodakis, “Wide residual networks,” in *Proc. BMVC*, 2016.
- [2] J. Ren *et al.*, “Balanced meta-softmax for long-tailed visual recognition,” in *Proc. NeurIPS*, 2020.
- [3] P. Izmailov, D. Podoprikin, T. Garipov, D. Vetrov, and A. G. Wilson, “Averaging weights leads to wider optima and better generalization,” in *Proc. UAI*, 2018.
- [4] S. Yun, D. Han, S. J. Oh, S. Chun, J. Choe, and Y. Yoo, “CutMix: Training strategy that makes use of sample pair-
ing,” in *Proc. ICCV*, 2019.
- [5] H. Zhang, M. Cisse, Y. N. Dauphin, and D. Lopez-Paz, “mixup: Beyond empirical risk minimization,” in *Proc. ICLR*, 2018.
- [6] D. Hendrycks *et al.*, “AugMix: A simple data processing method to improve robustness and uncertainty,” in *Proc. ICLR*, 2020.
- [7] T. DeVries and G. W. Taylor, “Improved regularization of convolutional neural networks with Cutout,” *arXiv:1708.04552*, 2017.